

ECSE 2500

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Topic: Covariance of two RVs

Conditional PDF of
random variable

□ Given a joint PDF/PMF/CDF of two RVs, what to determine if they are correlated?

→ Compute Covariance $\text{Cov}(X, Y) = 0$?

Example: $f_{X,Y}(x, y) = x + y$, if $x, y \in [0, 1]$.

What is $\text{Cov}(X, Y)$? Correlated?

Answer: Using the definition of Cov.

$$\text{Cov}(X, Y) = \underbrace{E[XY]} - \underbrace{E[X]} \underbrace{E[Y]}$$

$$\textcircled{1} E[XY]$$

$$= \int_0^1 \int_0^1 xy f_{XY}(x,y) dx dy$$

$$= \int_0^1 \int_0^1 xy (x+y) dx dy$$

$$= \int_0^1 \int_0^1 x^2 y dx dy + \int_0^1 \int_0^1 x y^2 dx dy$$

$$= \int_0^1 \frac{1}{3} x^3 y \Big|_{x=0}^1 dy + \int_0^1 \frac{1}{3} x y^3 \Big|_{y=0}^1 dx$$

$$= \frac{1}{3}$$

$$\textcircled{2} E[X]$$

$$= \int_0^1 x f_X(x) dx$$

$$= \int_0^1 x \underbrace{\int_0^1 f_{X,Y}(x,y) dy}_{f_X(x)} dx$$

$$= \int_0^1 \int_0^1 x(x+y) dx dy = \int_0^1 \left(\frac{1}{3} x^3 + \frac{1}{2} x^2 y \right) \Big|_0^1 dy = \frac{7}{12}$$

$$\textcircled{3} E[Y]$$

$$= \int_0^1 y f_Y(y) dy$$

$$= \int_0^1 y \underbrace{\int_0^1 f_{X,Y}(x,y) dx}_{f_Y(y)} dy$$

$$= \int_0^1 y \int_0^1 (x+y) dx dy = \frac{7}{12}$$

Then $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$

$$= \frac{1}{3} - \left(\frac{7}{12}\right)^2 \neq 0$$

They are correlated!

Note: By checking the Cov value above,

You may find $\text{Cov}(X, Y) < 0$.

This is in contrast with $\text{Var}(X) \geq 0$.

□ Correlation coefficients (e.g., ρ in Gaussian)

We often **normalize the covariance** of two RVs so that we pay less attention on their magnitude and immediately tell how related two RVs are.

Definition of **correlation coefficients** are

$$\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)} \sqrt{\text{Var}(Y)}} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

The nice thing about this definition is that

$$\rho_{X,Y} \in [-1, 1]$$

To see this point, we can consider

$$E\left[\left(\frac{X-\mu_x}{\sigma_x} \ominus \frac{Y-\mu_y}{\sigma_y}\right)^2\right] \geq 0$$

$$= E\left[\left(\frac{X-\mu_x}{\sigma_x}\right)^2 + \left(\frac{Y-\mu_y}{\sigma_y}\right)^2 - \frac{(X-\mu_x)(Y-\mu_y)}{\sigma_x \sigma_y}\right]$$

$$= E\left[\left(\frac{X-\mu_x}{\sigma_x}\right)^2\right] + E\left[\left(\frac{Y-\mu_y}{\sigma_y}\right)^2\right] - 2\rho_{xy}$$

$$= \frac{E[(X-\mu_x)^2]}{\sigma_x^2} + \frac{E[(Y-\mu_y)^2]}{\sigma_y^2} - 2\rho_{xy}$$

$$= 2 - 2\rho_{xy} \geq 0 \Rightarrow \rho_{xy} \leq 1$$

$$\Rightarrow \text{Show } E\left[\left(\frac{X-\mu_x}{\sigma_x} \oplus \frac{Y-\mu_y}{\sigma_y}\right)^2\right] \geq 0$$

$$\Rightarrow \rho_{xy} \geq -1$$

Q: How to interpret ρ_{XY} ?

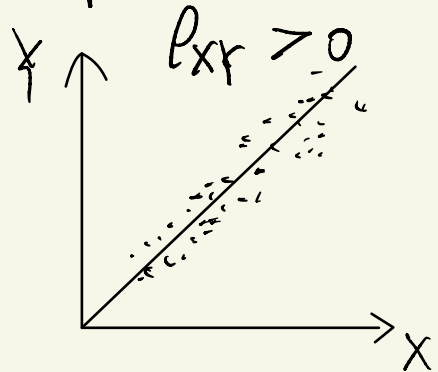
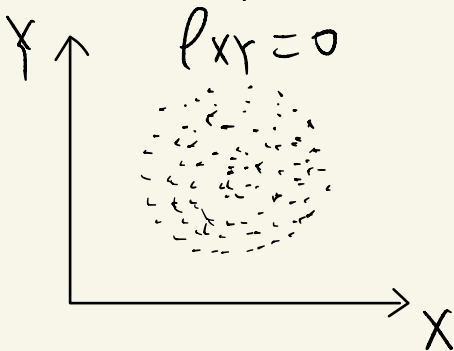
$\rho_{XY} = 0 \Rightarrow \text{Cov}(X, Y) = 0$, X, Y uncorrelated

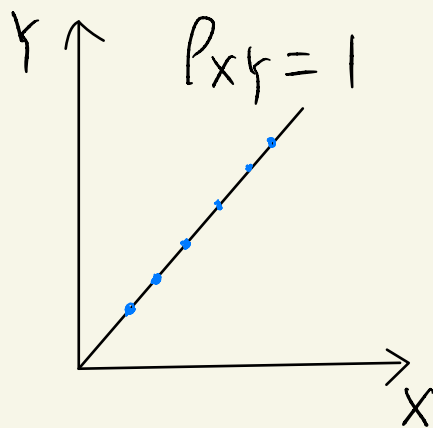
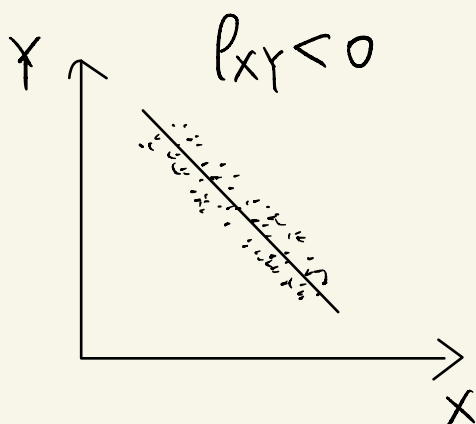
$\rho_{XY} = \pm 1 \Rightarrow Y$ is completely predictable given X

$\rho_{XY} > 0 \Rightarrow$ If X is above its mean, Y is likely also above its mean.

$\rho_{XY} < 0 \Rightarrow$ If X is above its mean, Y is likely below its mean.

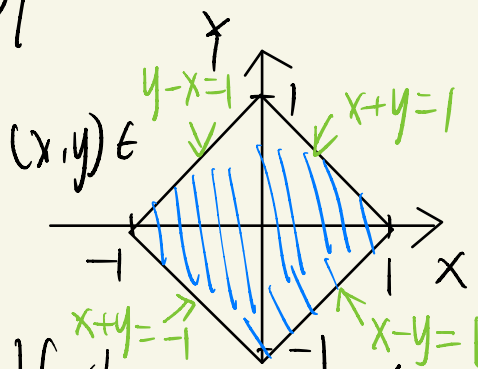
One way to visualize this is to look at a scatterplot of values of X and Y :





Example: Given joint PDF

$$f_{X,Y}(x,y) = \frac{1}{2}, \quad \text{if}$$



Q: What $\text{Cor}(X,Y)$? If they are correlated?

A:
$$E[XY] = \int_{-1}^{+1} \int_{-(1-x)}^{(1-x)} xy \cdot \frac{1}{2} dy dx$$

$$+ \int_{-1}^{+1} \int_{-(x+1)}^{(x+1)} xy \cdot \frac{1}{2} dy dx$$

$$= \int_0^1 \frac{1}{2} x \cdot \frac{1}{2} y^2 \Big|_{-(1-x)}^{1-x} dx + \int_{-1}^0 \frac{1}{2} x \cdot \frac{1}{2} y^2 \Big|_{-(x+1)}^{(x+1)} dx$$

$$= 0 + 0 = 0$$

$$E[X] = \int_{-1}^{+1} f_X(x) dx$$

$$f_X(x) = \int_{-1}^{+1} f_{X,Y}(x,y) dy$$

skip derivation

$$\stackrel{\downarrow}{=} 1 - |x|, \quad \forall -1 \leq x \leq 1$$

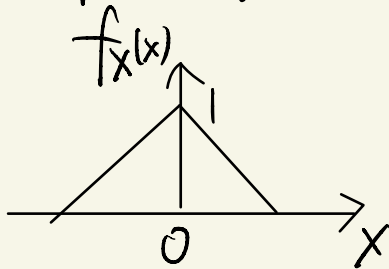
Likewise, we can find

$$f_Y(y) = 1 - |y|, \quad \forall -1 \leq y \leq 1$$

$$\Rightarrow E[X] = 0$$

Since PDF symmetric around 0.

$$E[Y] = 0$$



$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0$$

Note: X and Y are uncorrelated!

Q: Are they independent?

Necessary conditions of independent RVs

$$\checkmark f_{X,Y}(x,y) = f_X(x) f_Y(y)$$

$$F_{X,Y}(x,y) = F_X(x) F_Y(y)$$

Check $f_{X,Y}\left(\frac{3}{4}, \frac{1}{4}\right) = \frac{1}{2}$

$$f_X\left(\frac{3}{4}\right) \cdot f_Y\left(\frac{1}{4}\right) = \frac{1}{4} \cdot \frac{3}{4}$$

$$f_{X,Y}(x,y) \neq f_X(x) f_Y(y)$$

□ Conditional PDFs.

We have talked about conditional probability in terms of events in the sample space

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Also if B_i is a partition of sample space S

$$\left(\bigcup_{i=1}^n B_i = S, B_i \cap B_j = \emptyset \right)$$

then

$$P(A) = \sum_{i=1}^n P(A|B_i) P(B_i)$$

(Law of total probability)

Now we are interested in

$$P(Y \in A | X = x)$$

We first assume X is discrete, so $P(X = x_k) > 0$
for $S_X = \{x_1, x_2, \dots, x_k\}$.

If Y is also a discrete RV, we can look at their joint PMF and determine the conditional PMF, e.g.,

$$P(Y=y_j | X=x_k) = \frac{P(X=x_k \text{ and } Y=y_j)}{P(X=x_k)}$$

Using the marginal/joint PMF notations,

$$P_{Y|X}(y_j | x_k) = \frac{P_{X,Y}(x_k, y_j)}{P_X(x_k)} = \frac{\text{joint}}{\text{Marginal}}$$

Example: Recall the experiment of coin toss 3 times. Let

$X = \#$ of heads over 3 times

$Y = \text{position of first head}$

Outcome	HHH	HHT	HTH	THH	HTT	THT	TTH	TTT
X	3	2	2	2	1	1	1	0
Y	1	1	1	2	1	2	3	0

We can compute marginal PMFs

x	$P_X(x)$	y	$P_Y(y)$
0	$\frac{1}{8}$	0	$\frac{1}{8}$
1	$\frac{3}{8}$	1	$\frac{1}{2}$
2	$\frac{3}{8}$	2	$\frac{1}{4}$
3	$\frac{1}{8}$	3	$\frac{1}{8}$

We can compute joint PMF

(x, y)	$P_{X,Y}(x, y)$
$(0, 0)$	$\frac{1}{8}$
$(1, 1)$	$\frac{1}{8}$
$(1, 2)$	$\frac{1}{8}$
$(1, 3)$	$\frac{1}{8}$
$(2, 1)$	$\frac{1}{2} + \frac{1}{8} = \frac{5}{8}$
$(2, 2)$	$\frac{1}{8}$
$(3, 1)$	$\frac{1}{8}$
else	0

We can use joint PMFs and marginal PMFs to compute conditional PMFs.

$P(Y|X)$

$X \backslash Y$	0	1	2	3
0	$P(Y=0 X=0)$			
1	$P(Y=0 X=1)$			
2				
3				

$P(X|Y)$

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